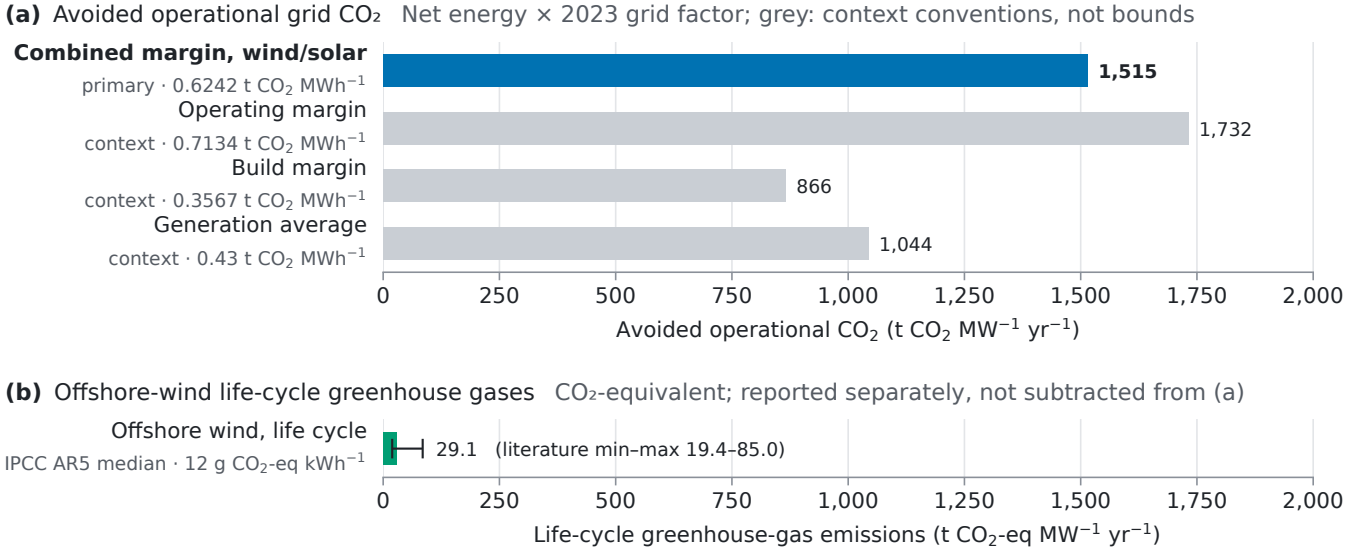
(a) Avoided operational grid CO₂
Net energy × 2023 grid factor; grey: context conventions, not bounds
Combined margin, wind/solar
primary · 0.6242 t CO2 MWh−1
1,515
Operating margin
context · 0.7134 t CO2 MWh−1
1,732
Build margin
context · 0.3567 t CO2 MWh−1
866
Generation average
context · 0.43 t CO2 MWh−1
1,044
0
250
500
750
1,000
1,250
1,500
1,750
2,000
Avoided operational CO2 (t CO2 MW−1 yr−1)
(b) Offshore-wind life-cycle greenhouse gases
CO₂-equivalent; reported separately, not subtracted from (a)
Offshore wind, life cycle
IPCC AR5 median · 12 g CO2-eq kWh−1
29.1 (literature min–max 19.4–85.0)
0
250
500
750
1,000
1,250
1,500
1,750
2,000
Life-cycle greenhouse-gas emissions (t CO2-eq MW−1 yr−1)